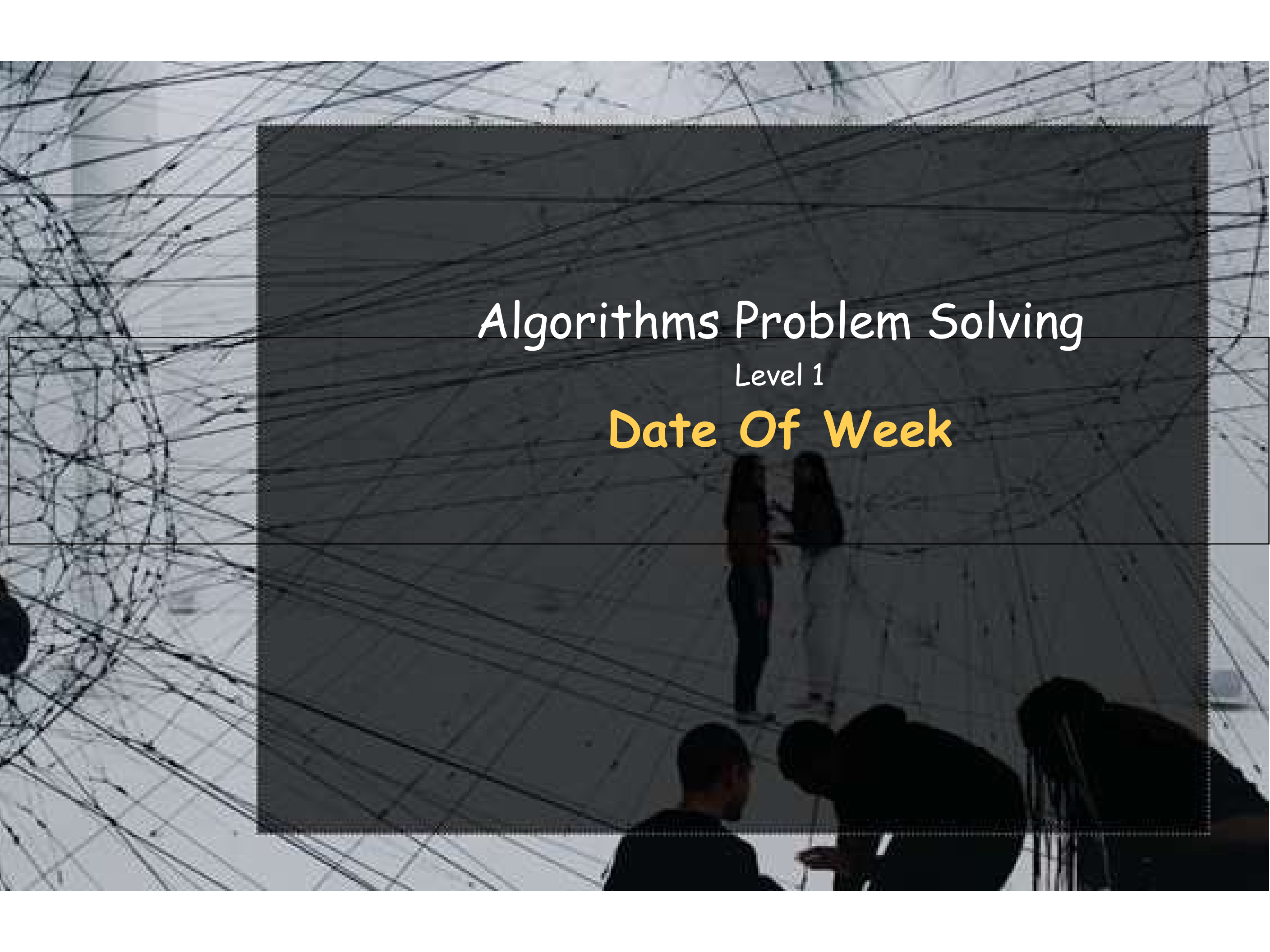


Algorithms Problem Solving

Level 1

Date Of Week



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Definition

- ◆ This program asks the user to enter a number from 1 to 7 and prints the corresponding day of the week.
- ◆ The program uses an enum to represent the days and a `switch` statement to convert the entered number into a day name.
- ◆ If the user enters a number outside 1-7, the program prints `Not a Week Day`.

Explanation

◆ ``enWeekDay`` is an enum representing the seven days:

1 = Sunday

2 = Monday

3 = Tuesday

4 = Wednesday

5 = Thursday

6 = Friday

7 = Saturday

◆ Because ``Sunday = 1`` is explicitly assigned, the following enum values automatically increase by 1.

◆ ``ShowWeekDayMenu()`` displays the available days and asks the user to enter a number.

◆ ``ReadWeekDay()`` reads an integer and converts it to the ``enWeekDay`` enum type.

◆ ``GetWeekDayName()`` uses a ``switch`` statement to match the enum value with the correct day name.

◆ The ``default`` case handles values that do not represent a valid day.

◆ In ``main()``, the result of ``ReadWeekDay()`` is passed directly to ``GetWeekDayName()``.

Steps in main()

1. Define the `enWeekDay` enum.
2. Assign numbers 1-7 to the days.
3. Display the week day menu.
4. Read the user's day number.
5. Convert the number to `enWeekDay`.
6. Use `switch` to identify the corresponding day.
7. Return the day name.
8. Print the result.

💡 Example

◆ If the user enters:

3

◆ The program matches `3` with:
`enWeekDay::Tuesday`

➡ Result

Today is Tuesday

Note

- ◆ An enum gives meaningful names to numeric values.
- ◆ Since `Sunday = 1`, the remaining days automatically receive values from 2 to 7.
- ◆ The `default` case handles invalid numbers such as 0 or 8.

⊘ Warning

◆ The ``break`` statements after ``return`` are unreachable because ``return`` already exits the function. They are unnecessary and can be removed.



Comparison

▨ Valid Day

- ✓ The entered number is between 1 and 7.
- ✓ It matches one of the enum values.
- ✓ The corresponding day name is returned.

▤ Invalid Day

- ✓ The entered number is outside 1-7.
- ✓ No `case` matches the value.
- ✓ The `default` case returns `Not a Week Day`.

✦✦ Summary

- ✓ An enum can represent related constant values with meaningful names.
- ✓ ``switch`` can select an action based on a specific value.
- ✓ ``default`` handles values that do not match any case.
- ✓ ``return`` exits the function, so a ``break`` after ``return`` is unnecessary.

Thank you 😊